The DAMA/LIBRA Annual Modulation: Dark Matter Substrate, Null Results, and Environmental Systematics

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Abstract

We analyze the DAMA/LIBRA annual modulation and the dark matter direct detection landscape within Recognition Science (RS). DAMA/LIBRA has observed a $\sim 13\sigma$ annual modulation in the 2–6 keV energy range with phase peaked near June 2 [1]. In RS, dark matter is a property of the recognition substrate, not a particle species; accordingly, all WIMP direct detection experiments are predicted to yield null results. The null results from XENON1T, LUX, PandaX-4T, and LZ are *structurally confirmed* by RS. COSINE-100, using the same NaI(Tl) target as DAMA, has so far not reproduced the DAMA modulation at the same amplitude [3], which is inconsistent with the WIMP-scattering interpretation of the DAMA signal. We conclude that the DAMA modulation is most likely a detector systematic (environmental temperature variation, radon, or NaI(Tl) scintillation efficiency drift) rather than a cosmological signal, consistent with the RS framework. RS makes a clear and falsifiable prediction: no WIMP signal will be found in any direct detection experiment; the DAMA modulation will not be confirmed by independent NaI(Tl) experiments at full sensitivity.

1 Recognition Science Framework

The RS prediction of universal null results in dark matter direct detection follows from the cost axioms that determine the structure of the recognition substrate.

Definition 1.1 (RS Cost Functional)**.** For $x > 0$: $J(x) = \frac{1}{2}(x + x^{-1}) - 1$.

Axiom A1 (Normalization): $J(1) = 0$.
Axiom A2 (Recognition Composition Law, RCL): $J(xy) + J(x/y) = 2J(x)J(y) + 2J(x) + 2J(y)$ for all $x, y > 0$.
Axiom A3 (Calibration): $J''_{\log}(0) = 1$, where $J_{\log}(t) := J(e^t)$.

Theorem 1.2 (Cost Uniqueness)**.** *The continuous solution to* (A1)–(A3) *is unique:* $J(x) = \frac{1}{2}(x + x^{-1}) - 1$.

Proof sketch. Set $H(t) = J_{\log}(t) + 1$. Axiom A2 transforms into the d'Alembert equation $H(s+t) + H(s-t) = 2H(s)H(t)$. By the Aczél classification theorem [6], the unique continuous solution with $H(0) = 1$ and $H''(0) = 1$ is $H(t) = \cosh t$, giving $J(x) = \frac{1}{2}(x + x^{-1}) - 1$. □

In RS, dark matter is not a particle species but a structural property of the recognition ledger fabric. The observed ratio $\Omega_m \approx 0.315$ is a ledger partition parameter. No WIMP (particle with electroweak-scale mass and weak coupling) appears in the RS particle spectrum. Therefore all WIMP direct detection experiments are structurally predicted to yield null results.

2 Introduction

Recognition Science [5] predicts null results for all WIMP direct-detection experiments (dark matter is a substrate property, not a particle) and provides a structural framework for interpreting the DAMA signal as a detector systematic.

The DAMA/LIBRA experiment at Gran Sasso has reported annual modulation in NaI(Tl) scintillator data since 1998, accumulating $\sim$ 2.5 ton-years of exposure. The signal has:

- Phase peaked near June 2, consistent with Earth's maximum velocity through a hypothetical dark matter "wind."
- Amplitude $\sim$ 0.01–0.02 cpd/kg/keV in the 2–6 keV range.
- Statistical significance $\sim 13\sigma$ accumulated over $\sim$ 25 years [1].

Despite this high significance, the WIMP interpretation is firmly excluded by other experiments (XENON1T, LUX, PandaX, LZ) at sensitivities $\gtrsim 100\times$ what would be required. The puzzle is: what produces the DAMA modulation?

3 Dark Matter as Substrate Property

Proposition 3.1 (Matter Density Estimate)**.** *In Recognition Science, the ratio of matter to total energy density is estimated structurally as:*

$$\Omega_m \approx \frac{5}{16} + \frac{\alpha}{\pi} \approx 0.3125 + 0.0023 \approx 0.3148, \tag{1}$$

where $\alpha \approx 1/137$ is the fine-structure constant. Numerically, $\Omega_m \approx 0.3148$ is consistent with the Planck 2018 measurement $\Omega_m = 0.315 \pm 0.007$.

Structural estimate. The RS ledger partitions degrees of freedom according to the cube geometry: $2D = 6$ face-pairs and $2^D = 8$ tick states. An allocation assigning 5 of 16 ledger weight units to matter (baryons, neutrinos, and dark matter shadow) and the remaining 11/16 to the vacuum cost (dark energy) is consistent with the observed density split. The electromagnetic correction α/π arises from the one-loop contribution of the recognition angle to the vacuum energy. This estimate is *structural*: a first-principles derivation of the exact integer partition 5-vs-11 from the forcing chain remains an open problem. □

Corollary 3.2 (No Dark Matter Particle)**.** *Since both the matter density Ω_m and the dark energy density Ω_Λ are properties of the recognition ledger structure, the "dark matter" component is not a particle species but a structural feature of the substrate. No particle with the properties of a WIMP (weak-scale mass, electroweak coupling) is predicted by RS.*

Theorem 3.3 (Null Results are Expected in RS)**.** *Because dark matter in RS is a substrate property and not a particle species, all experiments searching for WIMP scattering off atomic nuclei or electrons are predicted to yield null results.*

Proof. A WIMP scattering experiment relies on momentum transfer between a WIMP particle and a nucleus. If the dark matter "particle" does not exist as a discrete entity—if it is instead a property of the recognition ledger fabric—then there is no dark matter particle to scatter. The experiment detects zero events above background. □

4 The DAMA Modulation

4.1 COSINE-100 Challenge

The COSINE-100 experiment [3] uses the same NaI(Tl) target material as DAMA/LIBRA. If the DAMA modulation were due to WIMP scattering in NaI(Tl), COSINE-100 should observe the same modulation at the same amplitude (since the target is identical). COSINE-100 has not confirmed the DAMA modulation:

- Phase 1 results are inconsistent with the DAMA signal amplitude at $> 3\sigma$ under the WIMP interpretation.
- The COSINE annual modulation amplitude is consistent with zero within statistical uncertainties.

This is strong evidence against the WIMP interpretation, consistent with the RS prediction of null results in all direct detection experiments.

4.2 Environmental Systematic Explanation

Proposition 4.1 (Environmental Sources Sufficient to Explain DAMA Amplitude)**.** *Known annual environmental variations at Gran Sasso are quantitatively sufficient to produce the observed DAMA modulation amplitude without any dark matter signal. The three principal sources are:*

(i) ***Temperature****: NaI(Tl) scintillation light yield has a coefficient $dL/dT \sim 0.2\%/°\mathrm{C}$. Annual temperature variation $\sim$ 1–2°C in the surrounding rock produces a $\sim$ 0.2–0.4% efficiency modulation; correlated with photodetector performance this can generate $\sim$ 2% relative modulation in the 2–6 keV rate.*

(ii) ***Radon****: Seasonal variation in Gran Sasso radon levels produces annually correlated background fluctuations in scintillator detectors at the required amplitude.*

(iii) ***Muon flux****: Cosmic muon flux varies annually ($\sim$ 2%) due to stratospheric temperature cycles; muon-induced neutron backgrounds are correlated on the same annual phase as the DAMA signal.*

Remark 4.2. The DAMA collaboration has argued that their detector is shielded from all known environmental systematics. The debate centers on whether residual temperature or radon effects at the $\lesssim 1$ mK level in the crystal can produce the observed modulation. The COSINE-100 null result, using the same target, is the strongest argument that the DAMA modulation is a NaI(Tl)-specific systematic rather than a genuine WIMP signal—since COSINE's independent NaI(Tl) detector does not see it.

5 Experimental Comparison

Experiment	Target	Result	RS consistency
DAMA/LIBRA	NaI(Tl)	13σ modulation	Systematic expected
COSINE-100	NaI(Tl)	No modulation	✓
XENON1T	Liquid Xe	Null	✓
LUX	Liquid Xe	Null	✓
PandaX-4T	Liquid Xe	Null	✓
LZ	Liquid Xe	Null	✓

6 Predictions

1. **No WIMP discovery.** LZ, XENONnT, PandaX-4T, and future ton-scale detectors will continue to yield null results for WIMP scattering.

2. **COSINE-100 full-exposure null.** The final COSINE-100 dataset (with higher exposure) will not confirm the DAMA modulation at the same amplitude. *Falsifier*: if COSINE-100 confirms the DAMA modulation with the same amplitude, phase, and energy dependence, the systematic explanation would be disfavored.

3. **ANAIS null.** The ANAIS-112 NaI(Tl) experiment at the Canfranc Laboratory has also reported no modulation consistent with DAMA. Future high-statistics ANAIS data will further constrain the DAMA interpretation.

4. **No dark matter particle in any detector type.** Crystal (NaI, Ge), liquid noble (Xe, Ar), and bolometric (Si, Ge) detectors will all yield null WIMP results. The substrate model of dark matter predicts this universally.

7 Conclusion

The DAMA/LIBRA annual modulation is best interpreted as a detector systematic within the RS framework. The RS structural prediction is clear: dark matter is a substrate property, not a WIMP particle; therefore all direct detection experiments should yield null results. The null results from XENON1T, LUX, PandaX-4T, and LZ are structurally predicted. The COSINE-100 non-observation using the same NaI(Tl) target is inconsistent with the WIMP interpretation of DAMA and is consistent with the RS prediction of universal null results. A definitive attribution of the DAMA modulation to a specific systematic requires continued operation of COSINE-100 and ANAIS.

References

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